

## 11.4 The likelihood ratio test and its relatives

The  $F$  test was motivated in [Section 11.1](#) by a heuristic: compare the extra sum of squares with the noise level. General statistical theory offers other routes to a test, chiefly the likelihood ratio, the Wald statistic and the score statistic. In the normal linear model all three routes lead back to  $F$ . They differ only when they are calibrated by their common large-sample  $\chi^2$  approximation instead of the exact distribution, and then they can disagree about the same data. This section derives all three, shows how different the approximate versions can be, and then proves that the  $F$  test is the best test among those that respect the symmetries of [Section 11.2](#).

### 11.4.1 The likelihood ratio

For a hypothesis  $H_0$  about the parameter of a family of densities  $L(\boldsymbol{\theta}; \mathbf{y})$ , the **likelihood ratio** is

$$\Lambda(\mathbf{y}) = \frac{\sup_{\boldsymbol{\theta} \in H_0} L(\boldsymbol{\theta}; \mathbf{y})}{\sup_{\boldsymbol{\theta}} L(\boldsymbol{\theta}; \mathbf{y})},$$

and the likelihood ratio test rejects when  $\Lambda$  is small, that is, when the best explanation the hypothesis can offer is much worse than the best explanation overall. In the linear model both suprema are available in closed form from Theorem 7.3.1, and Exercise (A2) already carried out the computation for reduced models. The theorem records it for testable hypotheses, including the calibration.

#### Theorem 11.4.1 — The $F$ test is the likelihood ratio test

Assume Eq. 7.3.1 with  $r < n$ , and let  $H : \boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$  be testable with  $q = \text{rank}(\boldsymbol{\Lambda})$ . Let  $\text{SSE}_H = \min_{\boldsymbol{\Lambda}^\top \mathbf{b} = \mathbf{d}} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$  and suppose  $\text{SSE} > 0$ , which holds with probability one. Then:

- (a) under  $H$  the likelihood is maximized at any minimizer  $\hat{\boldsymbol{\beta}}_H$  of  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$  subject to  $\boldsymbol{\Lambda}^\top \mathbf{b} = \mathbf{d}$  (for  $\boldsymbol{\Lambda}$  of full column rank, the estimate of Proposition 11.3.3; compare Theorem 9.2.5), with  $\tilde{\sigma}_H^2 = \text{SSE}_H/n$ ;
- (b) the likelihood ratio is

$$\Lambda = \left( \frac{\text{SSE}}{\text{SSE}_H} \right)^{n/2} = \left( 1 + \frac{q F_H}{n - r} \right)^{-n/2}, \quad (11.4.1)$$

a strictly decreasing function of the statistic  $F_H$  of Eq. 11.3.1;

- (c) the likelihood ratio test of size  $\alpha$  rejects exactly when  $F_H > F_\alpha(q, n - r)$ .

The same holds for a reduced model  $\mathbf{E}(\mathbf{Y}) \in \mathcal{C}(\mathbf{X}_0)$ , with  $q = r - r_0$  and the statistic of Theorem 11.1.1.

*Proof.*

- (a) By Theorem 11.3.2(b),  $H$  holds iff  $\mathbf{E}(\mathbf{Y}) \in \mathbf{X}\mathbf{b}_0 + \mathcal{S}_0$ . Maximizing the likelihood over  $H$  is therefore maximizing it in the linear model  $\mathbf{Y} - \mathbf{X}\mathbf{b}_0 \sim N_n(\mathbf{m}, \sigma^2 \mathbf{I})$ ,

$\mathbf{m} \in \mathcal{S}_0$ . By Theorem 7.3.1 applied to that model (with any matrix whose columns span  $\mathcal{S}_0$ ), the maximizing mean is the least squares fit and the maximizing variance is its residual sum of squares divided by  $n$ . That fit is  $\mathbf{P}_{\mathcal{S}_0}(\mathbf{y} - \mathbf{X}\mathbf{b}_0)$ , so the maximizing mean of  $\mathbf{Y}$  is  $\mathbf{X}\mathbf{b}_0 + \mathbf{P}_{\mathcal{S}_0}(\mathbf{y} - \mathbf{X}\mathbf{b}_0)$ , the point of  $\mathbf{X}\mathbf{b}_0 + \mathcal{S}_0$  nearest to  $\mathbf{y}$ . As shown in the proof of Theorem 11.3.2(b), no rank condition on  $\mathbf{A}$  is needed for this. That point is  $\mathbf{X}\mathbf{b}$  exactly for the minimizers  $\mathbf{b}$  of  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$  subject to  $\mathbf{A}^\top \mathbf{b} = \mathbf{d}$ , so it is  $\mathbf{X}\hat{\boldsymbol{\beta}}_H$ , and its residual sum of squares is  $\text{SSE}_H$ . This needs  $\text{SSE}_H > 0$ , which follows from  $\text{SSE}_H \geq \text{SSE} > 0$ .

- (b) By Theorem 7.3.1 the two maxima are  $(2\pi e \text{SSE}/n)^{-n/2}$  and  $(2\pi e \text{SSE}_H/n)^{-n/2}$ . Their ratio is  $(\text{SSE}/\text{SSE}_H)^{n/2}$ , and  $\text{SSE}_H = \text{SSE} + \text{SS}_H = \text{SSE}\{1 + qF_H/(n-r)\}$  by Theorem 11.3.2(b).
- (c) By (b),  $\Lambda \leq k$  iff  $F_H \geq c$  for a constant  $c$  determined by  $k$ . By Theorem 11.3.2(c) the null distribution of  $F_H$  is  $F(q, n-r)$  at every point of  $H$ , so size  $\alpha$  requires  $c = F_\alpha(q, n-r)$ .

■

The large-sample theory of likelihood ratios (Wilks's theorem) would calibrate  $-2\log \Lambda = n\log\{1 + qF_H/(n-r)\}$  by the  $\chi^2(q)$  distribution. That calibration is only approximate. In the normal linear model the exact distribution is available through  $F_H$  and should be used.

## 11.4.2 Wald and score statistics

Two other large-sample recipes are in wide use. The **Wald statistic** measures the estimated departure  $\mathbf{A}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$  against its estimated covariance, both computed at the unrestricted maximum likelihood estimate. The **score statistic**, also called the Lagrange multiplier statistic, measures the gradient of the log-likelihood at the *restricted* maximum likelihood estimate against the information there. In the linear model, with  $\mathbf{A}$  of full column rank  $q$  and the maximum likelihood variance estimates  $\hat{\sigma}^2 = \text{SSE}/n$  and  $\tilde{\sigma}_H^2 = \text{SSE}_H/n$ , they are

$$\text{Wald} = \frac{(\mathbf{A}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{A}^\top \mathbf{G} \mathbf{A})^{-1} (\mathbf{A}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})}{\hat{\sigma}^2}$$

and

$$\text{Score} = \mathbf{U}^\top \left( \frac{\mathbf{X}^\top \mathbf{X}}{\tilde{\sigma}_H^2} \right)^- \mathbf{U} = \tilde{\sigma}_H^2 \mathbf{U}^\top (\mathbf{X}^\top \mathbf{X})^- \mathbf{U},$$

where  $\mathbf{U} = \mathbf{X}^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_H)/\tilde{\sigma}_H^2$ . Here  $\mathbf{U}$  is the gradient of the log-likelihood Eq. 7.3.1 with respect to  $\boldsymbol{\beta}$  at  $(\hat{\boldsymbol{\beta}}_H, \tilde{\sigma}_H^2)$ , and  $\mathbf{X}^\top \mathbf{X}/\tilde{\sigma}_H^2$  is the corresponding block of the Fisher information. The gradient with respect to  $\sigma^2$  vanishes at the restricted maximum, and the information matrix is block diagonal, because  $\mathbf{E}\{\mathbf{X}^\top (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})\} = \mathbf{0}$ . So the  $\sigma^2$  coordinates contribute nothing, and the formula is the full score statistic. The generalized inverse is harmless because  $\mathbf{U} \in \mathcal{C}(\mathbf{X}^\top \mathbf{X})$ .

### Proposition 11.4.2 — Wald, likelihood ratio and score

In the setting above, let  $u = \text{SS}_H/\text{SSE} = qF_H/(n-r)$ . Then

$$\text{Wald} = nu, \quad -2\log \Lambda = n\log(1+u), \quad \text{Score} = \frac{nu}{1+u}.$$

Each is a strictly increasing function of  $F_H$ , so the three tests, calibrated exactly, coincide with the  $F$  test. For every sample with  $SS_H > 0$ ,

$$\text{Wald} > -2 \log \Lambda > \text{Score}.$$

If each is compared with the upper  $\alpha$  point  $c$  of  $\chi^2(q)$ , the exact sizes are

$$\Pr\left\{F(q, n-r) > \frac{n-r}{q} u_c\right\}, \quad u_c = \frac{c}{n}, \quad e^{c/n} - 1, \quad \frac{c}{n-c} \quad (n > c)$$

for the Wald, likelihood ratio and score tests respectively.

*Proof.* With  $\hat{\sigma}^2 = \text{SSE}/n$ ,  $\text{Wald} = n \text{SS}_H / \text{SSE} = nu$  by Eq. 11.3.1. For the score,  $\mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}$ , so  $\mathbf{X}^\top (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}_H) = \mathbf{X}^\top \mathbf{X} (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)$  and

$$\mathbf{U}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{U} \cdot \tilde{\sigma}_H^2 = \frac{(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)^\top \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)}{\tilde{\sigma}_H^2} = \frac{\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)\|^2}{\tilde{\sigma}_H^2}.$$

By the proof of Proposition 11.3.3,  $\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)\|^2 = \text{SS}_H$ , so  $\text{Score} = n \text{SS}_H / \text{SSE}_H = nu/(1+u)$ . The likelihood ratio is Eq. 11.4.1. All three are increasing in  $u$ , which is proportional to  $F_H$ . For  $u > 0$ ,  $\log(1+u) = \int_0^u (1+t)^{-1} dt$  lies strictly between  $u/(1+u)$  and  $u$ , since the integrand lies strictly between  $(1+u)^{-1}$  and 1 on  $(0, u)$ . Finally, each statistic exceeds  $c$  iff  $u > u_c$ , with  $u_c$  as stated (for the score,  $nu/(1+u) > c$  iff  $u(n-c) > c$ , and it never exceeds  $c$  when  $n \leq c$ ), and  $u > u_c$  iff  $F_H > (n-r)u_c/q$ . ■

The inequality  $\text{Wald} \geq \text{LR} \geq \text{Score}$  is a general feature of linear restrictions in normal regression, and it has a practical consequence: with  $\chi^2$  critical values, the Wald test rejects most often and the score test least often, *on the same data*.

#### Example. (Three verdicts from one data set)

For the region test of Example (Do regions matter?),  $n = 50$ ,  $q = 3$  and  $n - r = 43$ . The three statistics are

| Statistic                           | Value | p-value from $\chi^2(3)$ |
|-------------------------------------|-------|--------------------------|
| Wald                                | 8.512 | 0.0365                   |
| Likelihood ratio, $-2 \log \Lambda$ | 7.861 | 0.0490                   |
| Score                               | 7.274 | 0.0637                   |
| $F = 2.440$ , exact                 |       | 0.0773                   |

With the 5% critical value 7.815 of  $\chi^2(3)$ , the Wald test rejects, the likelihood ratio test rejects barely, and the score test does not. The exact test, which all three approximate, does not reject. The disagreement is not a paradox. The approximate tests are different tests, and their true sizes differ from 0.05. With  $q = 3$  and  $r = 7$ , the formulas of [Proposition 11.4.2](#) give

| $n$                 | 15    | 30    | 50    | 100   | 400   |
|---------------------|-------|-------|-------|-------|-------|
| Wald                | 0.315 | 0.143 | 0.097 | 0.071 | 0.055 |
| Likelihood<br>ratio | 0.221 | 0.106 | 0.079 | 0.063 | 0.053 |
| Score               | 0.102 | 0.069 | 0.060 | 0.055 | 0.051 |

At the sample size of the example the nominal 5% Wald test has true size close to 10%. All three sizes approach 0.05 as  $n$  grows, but slowly when  $r$  is not small compared with  $n$ .

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statelcrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
codes = "SWWSWNSSSWMMMMSSNSNMMSMWNNWNSMMSWNNSMSSWNSWSMW"
region = np.array(list(codes))
y = data["murder"].to_numpy()
n = len(y)
covariates = np.column_stack([data["poverty"], data["single"], data["urban"]])
X0 = np.column_stack([np.ones(n), covariates])
D = np.column_stack([(region == g).astype(float) for g in "NSW"])
X = np.column_stack([X0, D])
r, q = 7, 3

def fit(Z):
    b, *_ = np.linalg.lstsq(Z, y, rcond=None)
    return b, np.sum((y - Z @ b) ** 2)

b, sse = fit(X)
b0, sse0 = fit(X0)
F = (sse0 - sse) / q / (sse / (n - r))

u = (sse0 - sse) / sse
wald = n * u
lr = n * np.log1p(u)
score = n * u / (1 + u)
crit = stats.chi2.ppf(0.95, q)
for name, stat in (("Wald", wald), ("LR", lr), ("score", score)):
    print(f"{name:5s} {stat:.3f} chi2 p = {stats.chi2.sf(stat, q):.4f}")
print(f"F      {F:.3f} exact p = {stats.f.sf(F, q, n - r):.4f}")
```

[Run this code online](#)

The lesson for other models is double. In the normal linear model none of the three approximations is needed, because the exact  $F$  test is available. In models without an exact theory,

generalized linear models among them (Chapter 34 onwards), the three statistics are the standard tools. Their disagreement in small samples is a warning that the  $\chi^2$  approximation is poor. The linear model suggests a remedy: dividing by  $q$  and referring to an  $F$  distribution with finite denominator degrees of freedom often calibrates better.

### 11.4.3 Invariance and optimality

Is the  $F$  test a *good* test? For a single constraint and a one-sided alternative, the  $t$  test of Section 11.6 has strong optimality properties. For  $q \geq 2$  no test is uniformly most powerful: the most powerful test against alternatives in one direction of the test space rejects for large values of the estimated departure *in that direction*, and it is a different test for each direction (Exercise (C1)). Some restriction on the class of tests is needed. The natural one is invariance. By Proposition 11.2.2, the problem does not change if the data are rescaled, shifted within the reduced model, or rotated within the test space and within the residual space. A test whose verdict changes under such operations would be favouring particular directions for no reason given by the problem. We show that among tests that do not, the  $F$  test is the most powerful, simultaneously against every alternative.

We need one property of the noncentral  $F$  family.

#### Lemma 11.4.3 — Monotone likelihood ratio of the noncentral $F$

For  $q, \nu \geq 1$  let  $f_\gamma$  be the density of  $F(q, \nu, \gamma)$ . For every  $\gamma > 0$ , the ratio  $f_\gamma(x)/f_0(x)$  is a strictly increasing function of  $x > 0$ .

*Proof.* Let  $V \sim \chi^2(\nu)$ , and for  $k = 0, 1, \dots$  let  $h_k$  be the density of  $(U_k/q)/(V/\nu)$ , where  $U_k \sim \chi^2(q + 2k)$  is independent of  $V$ . By Exercise (B4),  $B = U_k/(U_k + V) \sim \text{Beta}(a_k, b)$  with  $a_k = q/2 + k$  and  $b = \nu/2$ . The map  $B \mapsto x = \nu B/\{q(1 - B)\}$  is increasing, with inverse  $B = t(x) := qx/(\nu + qx)$ , and  $1 - t(x) = \nu/(\nu + qx)$ . Changing variables,

$$h_k(x) = \frac{t(x)^{a_k-1} \{1 - t(x)\}^{b-1}}{B(a_k, b)} \cdot \frac{q\nu}{(\nu + qx)^2}, \quad \text{so} \quad \frac{h_k(x)}{h_0(x)} = \frac{B(q/2, \nu/2)}{B(q/2 + k, \nu/2)} t(x)^k = c_k t(x)^k,$$

where  $B$  is the beta function and  $c_k > 0$ . By Theorem 4.1.2(d),  $U \sim \chi^2(q, \gamma)$  has the law of  $U_K$  with  $K \sim \text{Poisson}(\gamma/2)$  independent of the  $U_k$  and  $V$ . Conditioning on  $K$ ,  $F = (U/q)/(V/\nu)$  has density  $f_\gamma(x) = \sum_{k \geq 0} p_k h_k(x)$  with  $p_k = e^{-\gamma/2} (\gamma/2)^k / k!$ , and  $f_0 = h_0$ . Hence

$$\frac{f_\gamma(x)}{f_0(x)} = \sum_{k \geq 0} p_k c_k t(x)^k.$$

Every coefficient is positive when  $\gamma > 0$ , so the series is strictly increasing in  $t$ , and  $t(x)$  is strictly increasing in  $x$ . ■

A test is a measurable function  $\phi$  of the data with values in  $[0, 1]$ , the probability of rejecting. It is **invariant** if  $\phi(g(\mathbf{y})) = \phi(\mathbf{y})$  for all  $\mathbf{y}$  and all  $g$  in the set  $\mathcal{G}$  of maps

$$g(\mathbf{y}) = a\mathbf{H}\mathbf{y} + \mathbf{v}, \quad a > 0, \quad \mathbf{v} \in \mathcal{C}(\mathbf{X}_0), \quad \mathbf{H} \text{ orthogonal with } \mathbf{H}\mathcal{C}(\mathbf{X}_0) = \mathcal{C}(\mathbf{X}_0), \quad \mathbf{H}\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{X}).$$

For a testable hypothesis with  $\mathbf{d} \neq \mathbf{0}$ , the same applies to the data  $\mathbf{y} - \mathbf{X}\mathbf{b}_0$ , with  $\mathcal{C}(\mathbf{X}_0)$  replaced by  $\mathcal{S}_0$ .

### Theorem 11.4.4 — The $F$ test is uniformly most powerful invariant

Assume Eq. 7.3.1 and the setting of Definition 11.1.1. Let  $\phi$  be an invariant test with  $E\phi(\mathbf{Y}) \leq \alpha$  at every point of  $H_0$ . Then at every point  $(\beta, \sigma^2)$  of the full model,

$$E_{\beta, \sigma^2} \phi(\mathbf{Y}) \leq \Pr\{F(q, n-r, \gamma) > F_\alpha(q, n-r)\},$$

with  $\gamma$  given by Eq. 11.1.2. The right side is the power of the  $F$  test, which is itself invariant and of size  $\alpha$ .

*Proof.* Use the canonical coordinates  $\mathbf{z}_k = \mathbf{Q}_k^\top \mathbf{y}$  of Proposition 11.2.3, and let  $\mathcal{Y} = \{\mathbf{y} : \mathbf{z}_2 \neq \mathbf{0}\}$ , a set of probability one on which  $F$  is defined.

*Step 1: an invariant test depends on  $\mathbf{y}$  only through  $F$ .* Let  $\mathbf{y}, \mathbf{y}' \in \mathcal{Y}$  with  $F(\mathbf{y}) = F(\mathbf{y}')$ . Put  $a = \|\mathbf{z}'_2\|/\|\mathbf{z}_2\| > 0$ . Equality of the  $F$  values gives  $\|\mathbf{z}'_1\| = a\|\mathbf{z}_1\|$ . Choose orthogonal matrices  $\mathbf{O}_1$  ( $q \times q$ ) and  $\mathbf{O}_2$   $((n-r) \times (n-r))$  with  $\mathbf{O}_1(a\mathbf{z}_1) = \mathbf{z}'_1$  and  $\mathbf{O}_2(a\mathbf{z}_2) = \mathbf{z}'_2$ . This is possible because the vectors in each pair have equal lengths, and if  $\mathbf{z}_1 = \mathbf{0}$  then  $\mathbf{z}'_1 = \mathbf{0}$  and  $\mathbf{O}_1 = \mathbf{I}$  will do. Put

$$\mathbf{H} = \mathbf{Q}_0\mathbf{Q}_0^\top + \mathbf{Q}_1\mathbf{O}_1\mathbf{Q}_1^\top + \mathbf{Q}_2\mathbf{O}_2\mathbf{Q}_2^\top, \quad \mathbf{v} = \mathbf{Q}_0(\mathbf{z}'_0 - a\mathbf{z}_0).$$

$\mathbf{H}$  is orthogonal, maps  $\mathcal{C}(\mathbf{X}_0) = \mathcal{C}(\mathbf{Q}_0)$  and  $\mathcal{C}(\mathbf{X}) = \mathcal{C}([\mathbf{Q}_0, \mathbf{Q}_1])$  onto themselves, and  $\mathbf{v} \in \mathcal{C}(\mathbf{X}_0)$ . The canonical coordinates of  $a\mathbf{H}\mathbf{y} + \mathbf{v}$  are  $a\mathbf{z}_0 + \mathbf{z}'_0 - a\mathbf{z}_0 = \mathbf{z}'_0$ ,  $a\mathbf{O}_1\mathbf{z}_1 = \mathbf{z}'_1$  and  $a\mathbf{O}_2\mathbf{z}_2 = \mathbf{z}'_2$ , so  $a\mathbf{H}\mathbf{y} + \mathbf{v} = \mathbf{y}'$  and  $\phi(\mathbf{y}') = \phi(\mathbf{y})$ . Now fix, for each  $f \geq 0$ , the point  $\mathbf{y}_f = \sqrt{qf/(n-r)}\mathbf{Q}_1\mathbf{e}_1 + \mathbf{Q}_2\mathbf{e}_1$ , where  $\mathbf{e}_1$  denotes a first coordinate vector, and put  $\psi(f) = \phi(\mathbf{y}_f)$ . Then  $F(\mathbf{y}_f) = f$ , so  $\phi(\mathbf{y}) = \psi(F(\mathbf{y}))$  for every  $\mathbf{y} \in \mathcal{Y}$ . The map  $f \mapsto \mathbf{y}_f$  is continuous, so  $\psi$  is measurable.

*Step 2: reduction to one dimension.* By Step 1 and Theorem 11.1.1(b),  $E\phi(\mathbf{Y}) = E\psi(F) = \int_0^\infty \psi(x)f_\gamma(x)dx$ , with  $f_\gamma$  the density of  $F(q, n-r, \gamma)$ . The hypothesis is  $\gamma = 0$  (Theorem 11.1.1(c)), so the size condition says  $\int \psi f_0 \leq \alpha$ .

*Step 3: Neyman–Pearson.* Fix  $\gamma > 0$ , let  $c = F_\alpha(q, n-r)$ ,  $\phi^*(x) = 1\{x > c\}$  and  $k = f_\gamma(c)/f_0(c)$ . By Lemma 11.4.3,  $f_\gamma(x) - kf_0(x)$  is positive for  $x > c$  and negative for  $x < c$ . Since  $\phi^* - \psi \geq 0$  where  $x > c$  and  $\phi^* - \psi \leq 0$  where  $x < c$ , the product  $(\phi^* - \psi)(f_\gamma - kf_0)$  is nonnegative everywhere. Integrating,

$$\int (\phi^* - \psi)f_\gamma \geq k \int (\phi^* - \psi)f_0 = k\left(\alpha - \int \psi f_0\right) \geq 0.$$

So  $E\phi(\mathbf{Y}) = \int \psi f_\gamma \leq \int \phi^* f_\gamma$ , which is the power of the  $F$  test. For  $\gamma = 0$  the inequality is the size condition. The  $F$  test is invariant by Proposition 11.2.2. ■

The theorem says more than “the  $F$  test is good”. It says that once one agrees that no direction in the test space and no scale deserves special treatment, the choice of test is forced: every invariant test is a function of  $F$  (Step 1), and among functions of  $F$  the upper tail is best against every alternative at once. The likelihood ratio principle arrives at the same test without

any appeal to optimality, which is reassuring but not a proof. In other models the likelihood ratio test need not be optimal in any sense.

*Remark* (What invariance costs). Invariance is a restriction, and the  $F$  test is not the most powerful test against every alternative. If one contrast in the test space is of special interest, the  $t$  test for that contrast is more powerful against departures in its direction and less powerful against others (Section 11.6). A Bayesian analysis with a prior concentrated on some directions behaves similarly. Invariance expresses the absence of such preferences. It is appropriate for an omnibus question like “do the regions differ?”, and inappropriate when the scientific question singles out a direction.

## 11.4.4 Exercises

### 11.4.4.1 A. Check your understanding

#### Exercise A1.

From  $F = 2.440$ ,  $q = 3$ ,  $n = 50$  and  $r = 7$ , compute the three statistics of Proposition 11.4.2 and check them against Example (Three verdicts from one data set).

### 11.4.4.2 B. Practice

#### Exercise B1.

Fix  $q$  and  $r$ . Show that under  $H$ , as  $n \rightarrow \infty$ ,  $qF_H$  converges in distribution to  $\chi^2(q)$ , and that the differences Wald  $- qF_H$ ,  $-2\log \Lambda - qF_H$  and Score  $- qF_H$  converge to zero in probability. Deduce that the sizes in Example (Three verdicts from one data set) tend to 0.05.

*Solution.* Write  $qF_H = (SS_H/\sigma^2)/(s^2/\sigma^2)$ . The numerator is  $\chi^2(q)$  for every  $n$ , and  $s^2/\sigma^2 \rightarrow 1$  in probability (Exercise (B1)), so  $qF_H \rightarrow \chi^2(q)$  in distribution. With  $u = qF_H/(n - r)$ , Wald  $- qF_H = qF_H\{n/(n - r) - 1\} = qF_H r/(n - r) \rightarrow 0$  in probability. For the others,  $nu = O_p(1)$ , so  $u \rightarrow 0$ , and  $n \log(1 + u) - nu = nu\{\log(1 + u)/u - 1\}$  and  $nu/(1 + u) - nu = -nu^2/(1 + u)$  both tend to zero. Each statistic therefore converges in distribution to  $\chi^2(q)$ , whose upper 5% point is the critical value used, so the sizes converge to 0.05.

#### Exercise B2.

Let  $\hat{\epsilon}_H = \mathbf{y} - \mathbf{X}\hat{\beta}_H$  be the residual vector of the restricted fit. Show that the score statistic equals  $n\|\mathbf{M}\hat{\epsilon}_H\|^2/\|\hat{\epsilon}_H\|^2$ , which is  $n$  times the uncentred  $R^2$  from regressing  $\hat{\epsilon}_H$  on  $\mathbf{X}$ . Explain why this “ $nR^2$ ” recipe needs only the restricted fit.

*Solution.*  $\mathbf{M}\hat{\epsilon}_H = \mathbf{M}\mathbf{y} - \mathbf{X}\hat{\beta}_H = \mathbf{X}(\hat{\beta} - \hat{\beta}_H)$ , whose squared length is  $SS_H$  (proof of Proposition 11.4.2), and  $\|\hat{\epsilon}_H\|^2 = SSE_H$ . So  $n\|\mathbf{M}\hat{\epsilon}_H\|^2/\|\hat{\epsilon}_H\|^2 = nSS_H/SSE_H = \text{Score}$ . The uncentred  $R^2$  of a regression of  $\hat{\epsilon}_H$  on  $\mathbf{X}$  is  $\|\mathbf{M}\hat{\epsilon}_H\|^2/\|\hat{\epsilon}_H\|^2$ . Computing it requires the restricted residuals and one auxiliary regression, but not  $\hat{\beta}$  or the unrestricted model’s residual sum of squares as such. This is why score tests are popular when the

full model is hard to fit, as in tests for omitted nonlinear terms or for heteroscedasticity (Chapter 21).

### Exercise B3.

Suppose  $\sigma^2 = \sigma_0^2$  is known. Show that the likelihood ratio test of a reduced model rejects for large  $(\text{SSE}_0 - \text{SSE})/\sigma_0^2$ , the statistic of Exercise (B1), and that  $-2\log \Lambda$  is then exactly  $\chi^2(q)$  under  $H_0$ .

#### 11.4.4.3 C. Going deeper

### Exercise C1.

In the canonical form with  $\sigma^2$  known and  $q \geq 2$ , consider testing  $\boldsymbol{\eta}_1 = \mathbf{0}$  against the single alternative  $\boldsymbol{\eta}_1 = \mathbf{a} \neq \mathbf{0}$ . Show by the Neyman–Pearson lemma that the most powerful level- $\alpha$  test rejects when  $\mathbf{a}^\top \mathbf{z}_1 > \sigma \|\mathbf{a}\| z_\alpha$ , where  $z_\alpha$  is the upper  $\alpha$  point of  $N(0,1)$ . Show that this test has power below  $\alpha$  against  $-\mathbf{a}$ , and conclude that no uniformly most powerful test of  $\boldsymbol{\eta}_1 = \mathbf{0}$  exists.

*Solution.* The likelihood ratio of  $N(\mathbf{a}, \sigma^2 \mathbf{I})$  to  $N(\mathbf{0}, \sigma^2 \mathbf{I})$  at  $\mathbf{z}_1$  is  $\exp\{(\mathbf{a}^\top \mathbf{z}_1 - \|\mathbf{a}\|^2/2)/\sigma^2\}$ , increasing in  $\mathbf{a}^\top \mathbf{z}_1$ . Under the null,  $\mathbf{a}^\top \mathbf{z}_1 \sim N(0, \sigma^2 \|\mathbf{a}\|^2)$ , which gives the critical value, and the Neyman–Pearson argument of Step 3 of the proof of [Theorem 11.4.4](#) shows that this test is most powerful. Against  $-\mathbf{a}$ ,  $\mathbf{a}^\top \mathbf{z}_1 \sim N(-\|\mathbf{a}\|^2, \sigma^2 \|\mathbf{a}\|^2)$ , so the power is  $\Pr\{Z > z_\alpha + \|\mathbf{a}\|/\sigma\} < \alpha$ . A uniformly most powerful test would have to be most powerful against both  $\mathbf{a}$  and  $-\mathbf{a}$ . The most powerful tests against these two alternatives are essentially unique and different, so no single test can be both.

### Exercise C2.

Extend [Lemma 11.4.3](#): show that for  $0 \leq \gamma_1 < \gamma_2$  the ratio  $f_{\gamma_2}(x)/f_{\gamma_1}(x)$  is increasing in  $x$ . (Write both densities as power series in  $t(x)$  with coefficients  $a_k(\gamma)$ , and show that  $a_k(\gamma_2)/a_k(\gamma_1)$  increases with  $k$ .) Deduce that the  $F$  test is also uniformly most powerful invariant for testing  $\gamma \leq \gamma_1$  against  $\gamma > \gamma_1$ , with a noncentral critical value. Such tests appear in equivalence testing.